Trey Kirscht

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Summary

Electrical Engineering Co-op student specializing in embedded systems, hardware/firmware integration, mixed-signal prototyping, and bench validation. Experienced with low-level C/assembly, Python tooling, PCB-based hardware, instrumentation, power/control systems, and RTOS-style concurrency. Built and validated prototypes spanning intraspinal stimulation hardware, rectifier control, preemptive AVR RTOS development, waveform generation, and custom microcontroller programming tools.

Education

University of Alberta Expected 2026
Bachelor of Science in Electrical Engineering, Co-op *Edmonton, AB*

Technical Skills

Programming	C, C++, Python, MATLAB, VBA, AVR/PIC Assembly, ARM Assembly
Embedded	Microcontroller firmware, timers/interrupts, RTOS primitives, context switching, UART, SPI, I2C, ADC/DAC, hardware/software debugging
Hardware	Schematic capture, PCB layout, mixed-signal prototyping, power electronics fundamentals, analog measurement, soldering, board bring-up
Lab/Test	Oscilloscopes, multimeters, vector network analyzers, signal generators, logic analyzers, bench validation
Tools	Altium Designer, LTspice, PathWave ADS, Cadence Virtuoso, Ghidra, Git, Digilent WaveForms SDK

Engineering Experience

University of Alberta Research Lab 2026
Research Engineering Co-op / Hardware-Firmware Prototype Development *Edmonton, AB*

- Designed and validated a routed 48-channel intraspinal stimulation prototype using eight active stimulation lanes, analog current-source/sink output stages, crosspoint electrode routing, muxed measurement, and AD3 waveform control.
- Breadboarded and tested a mixed-signal output stage, validating biphasic current pulses, command-capacitor timing, measurement-path behavior, and compliance limits under representative loads.
- Developed Python/WaveForms tooling for hardware-timed analog/digital waveform generation, pulse sequencing, measurement capture, calibration workflows, and test UI visualization.
- Designed the full prototype PCB, including power regulation, AD3 control interface, output stages, analog routing, measurement muxing, common-return switching, status indicators, and staged bring-up planning.

Seer Innovations Inc. Jan 2024 – Sep 2025
Engineering Support Specialist *Canada*

- Contributed firmware, testing, and integration work for a cathodic protection rectifier replacement using controllable switch-mode supplies, measurement, logging, remote monitoring, and operator tooling.
- Produced technical reports and engineering deliverables for inspection/corrosion-protection projects, including graphs, anomaly tables, data summaries, and appendices.
- Built internal tools for survey data processing, CSV-to-KMZ generation, field-use mapping deliverables, and VBA-based workflow automation.

Weir Concepts Jun 2022 – Sep 2023
Shop Technician *Edmonton, AB*

- Operated plasma cutting, CNC, and saw equipment; assembled and pressure-tested mechanical and hydraulic systems to API standards.

Selected Projects

Cathodic Protection Rectifier Controller Unit 2025 – 2026
Capstone – Embedded / Firmware Lead

- Developed Particle Boron firmware for a retrofit rectifier controller with zero-cross synchronized SCR phase control, RMS/DC measurement, calibration, and closed-loop voltage control.
- Built cloud telemetry, diagnostics, waveform capture, history logging, and a Flask-based web UI for testing, validation, and control on real rectifier hardware.

BadRTOS / PIC TinyRTOS Runtime 2026
Preemptive AVR RTOS and PIC16F84A Cooperative Runtime

- Designed and hardware-validated a preemptive RTOS for ATmega328P in C/AVR assembly with manual context switching, separate task stacks, timer interrupt scheduling, task delays, and priority scheduling.
- Implemented semaphores, mutexes with priority inheritance, mailbox IPC, stack high-water monitoring, non-yielding task preemption, GPIO instrumentation, and UART diagnostics.
- Built a PIC16F84A assembly precursor with a Timer0 tick, countdown-based cooperative task timing, heartbeat modes, debounced button input, software PWM, and bit-banged UART telemetry.
- Public repository: github.com/trey109231/badRTOS.

OpenAD3Flash 2026

Analog Discovery 3 Microcontroller Programming and Flash Tooling

- Built Python/WaveForms tools using a Digilent Analog Discovery 3 for AVR ISP programming, PIC16 high-voltage programming, EEPROM dumping, SPI flash dumping, and read-only glitch experiments.
- Implemented low-level bit-banged protocols, target-power control, signature reads, Intel HEX parsing, flash/verify workflows, and non-destructive diagnostic tooling. Repository: github.com/trey109231/OpenAD3Flash.

Embedded Waveform, Capture, and Distributed Compute Tools 2024 – 2026

AVR / STM32 / ATtiny Firmware Projects

- Developed AVR waveform-generation firmware with timer-driven output, external DAC control, LCD output, joystick/keypad input, and configurable waveform parameters.
- Built an STM32 timer/DMA waveform-generation and ADC-capture workflow with UART commands, circular DMA, binary capture export, and Python plotting.
- Built a five-node ATtiny84A Mandelbrot cluster with a shared SPI-style bus, fixed-point worker firmware, and Python host-side dynamic row scheduling.

Additional Technical Projects

RF Amplifier Design and Test 2025 *Course Project*

Designed and tested a 2.3 GHz narrowband RF amplifier in PathWave ADS using Smith-chart matching, S-parameter simulation, PCB layout, VNA testing, and matching-network tuning.

Multilayer Device Fabrication and Characterization 2025 *Course Project*

Fabricated and characterized a multilayer silicon microdevice using PECVD silicon nitride, aluminum sputtering, photolithography, wet etching, and electrical probing.